

SIMATS ENGINEERING
ASSESSMENT TOOL I – EXPERIMENTS

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : Name:

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4) Activity Tool : Experiments

Weightage: 20%

Code of Conduct:

I _P.Sunayana (192424303)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Sunayana

Criteria	Understanding of Concepts 25%	Experimental Design 30%	Use of Tools 20%	Analysis of Results 15%	Documents 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

DSA06 – DATA HANDLING AND VISUALIZATION

CO3_AT1_Assessment Tool: EXPERIMENT

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162,

171, 184, 205, 249, 318 **Tasks:**

1. Construct an ECDF and a normal Q-Q plot.
2. Use both plots to evaluate the normality claim.
3. Identify evidence of skewness and/or extreme observations.
4. Recommend percentile-based limits or mean $\pm$ standard-deviation limits, with justification.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation

Code:

```
library(ggplot2)

response_time <- c(
  118, 121, 125, 127, 129,
  131, 134, 136, 138, 141,
  144, 147, 151, 156, 162,
  171, 184, 205, 249, 318
)

ecdf_data <- data.frame(response_time)

ggplot(ecdf_data, aes(x = response_time)) +
  stat_ecdf(geom = "step") +
```

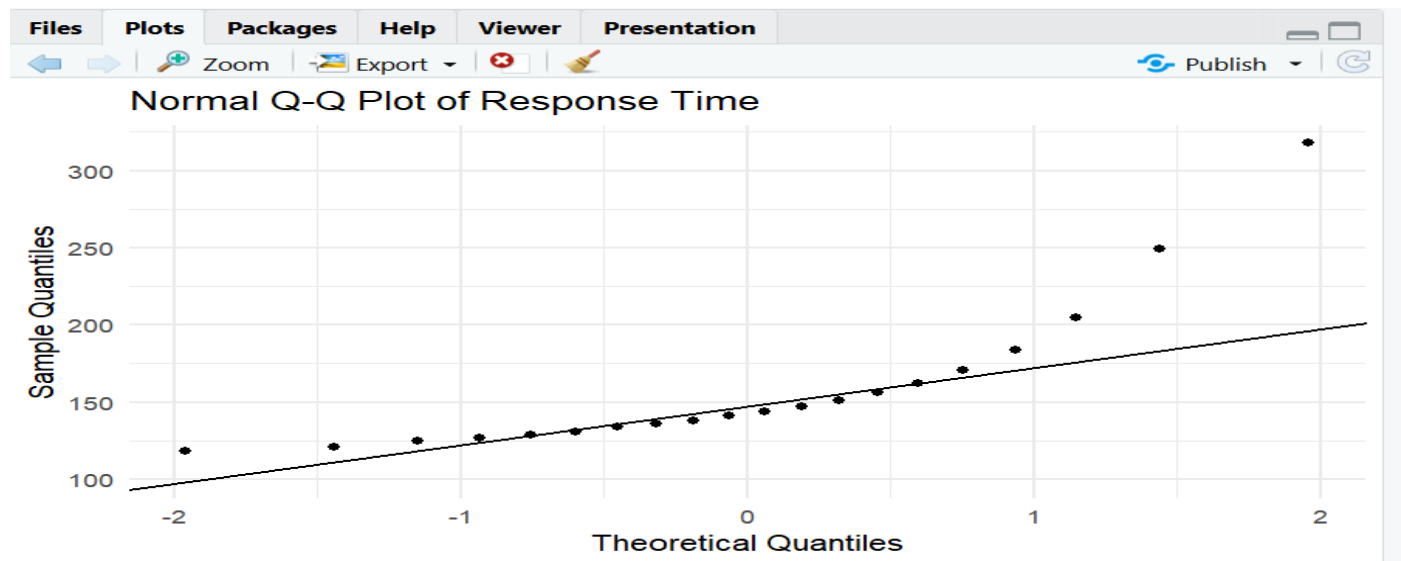
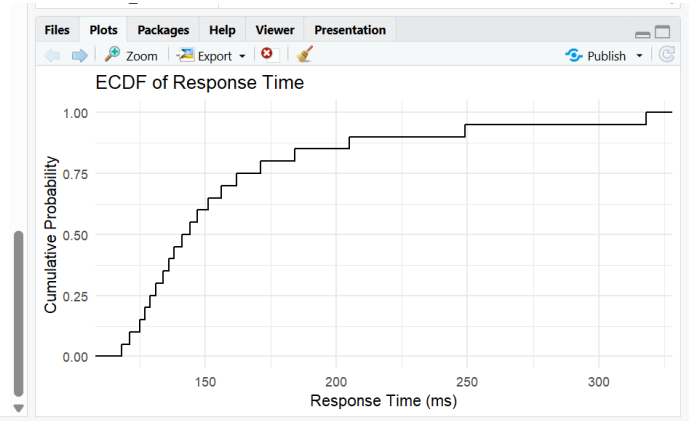
```

labs(
  title = "ECDF of Response Time",
  x = "Response Time (ms)",
  y = "Cumulative Probability"
) +
theme_minimal()
ggplot(ecdf_data, aes(sample = response_time)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Normal Q-Q Plot of Response Time",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles" ) +
  theme_minimal()
mean_value <- mean(response_time)
median_value <- median(response_time)
sd_value <- sd(response_time)
skewness_value <- mean((response_time - mean_value)^3) / sd_value^3
cat("Mean:", mean_value, "\n")
cat("Median:", median_value, "\n")
cat("Standard Deviation:", sd_value, "\n")
cat("Approximate Skewness:", skewness_value, "\n")
percentile_limits <- quantile(
  response_time,
  probs = c(0.05, 0.95))

```

```
print(percentile_limits)
```

```
> cat("Mean:", mean_value, "\n")
Mean: 159.35
> cat("Median:", median_value, "\n")
Median: 142.5
> cat("Standard Deviation:", sd_value, "\n")
Standard Deviation: 49.00191
> cat("Approximate Skewness:", skewness_value, "\n")
Approximate Skewness: 1.921656
>
> # -----
> # 4. Percentile-Based Limits
> # -----
> percentile_limits <- quantile(
+   response_time,
+   probs = c(0.05, 0.95)
+ )
> print(percentile_limits)
5%    95%
120.85 252.45
> |
```



1. ECDF

The ECDF shows that most response times are concentrated approximately between **118 ms and 205 ms**. However, the observations **249 ms and 318 ms** create a long upper tail.

2. Q-Q Plot

If the data were normally distributed, the points would approximately follow a straight line.

Here, the upper observations are expected to deviate noticeably from the straight reference line, especially **249 ms and 318 ms**.

3. Evidence of Skewness and Extreme Observations

- The distribution is **right-skewed (positively skewed)**.
- **318 ms** is an extreme high observation.
- **249 ms** is also unusually high compared with most observations.
- The mean is likely to be greater than the median because of the large upper values.

4. Recommendation

Use **percentile-based limits** rather than:

Mean $\pm$ Standard Deviation

Justification

The data is not approximately symmetric and contains extreme high values. Mean and standard deviation are sensitive to such observations.

Therefore, percentile-based monitoring limits such as the **5th percentile and 95th percentile** are more robust.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Production Line	Cycle Times (minutes)
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. Create a visualization comparing all four distributions.
2. Construct bean plots or an equivalent distribution visualization along the vertical axis.
3. Determine which line has greatest variability and which has the most concentrated distribution.
4. Explain why comparing only means is insufficient.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability

```
L1 <- c(42, 44, 45, 46, 47, 48, 49, 50, 51, 53)
```

```
L2 <- c(38, 41, 44, 47, 50, 53, 56, 59, 62, 65)
```

```
L3 <- c(45, 46, 46, 47, 47, 48, 48, 49, 49, 50)
```

```
L4 <- c(40, 41, 42, 43, 44, 45, 48, 55, 61, 72)
```

```
boxplot(
```

```
  L1, L2, L3, L4,
```

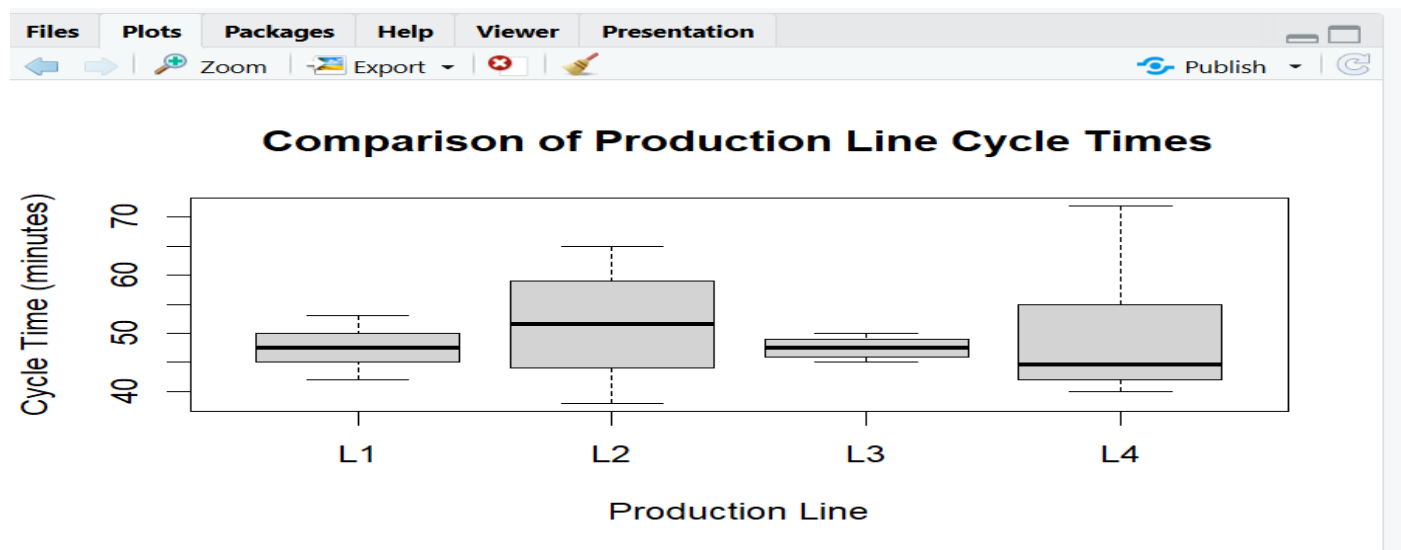
```
  names = c("L1", "L2", "L3", "L4"),
```

```
  main = "Comparison of Production Line Cycle Times",
```

```
  xlab = "Production Line",
```

```
  ylab = "Cycle Time (minutes)"
```

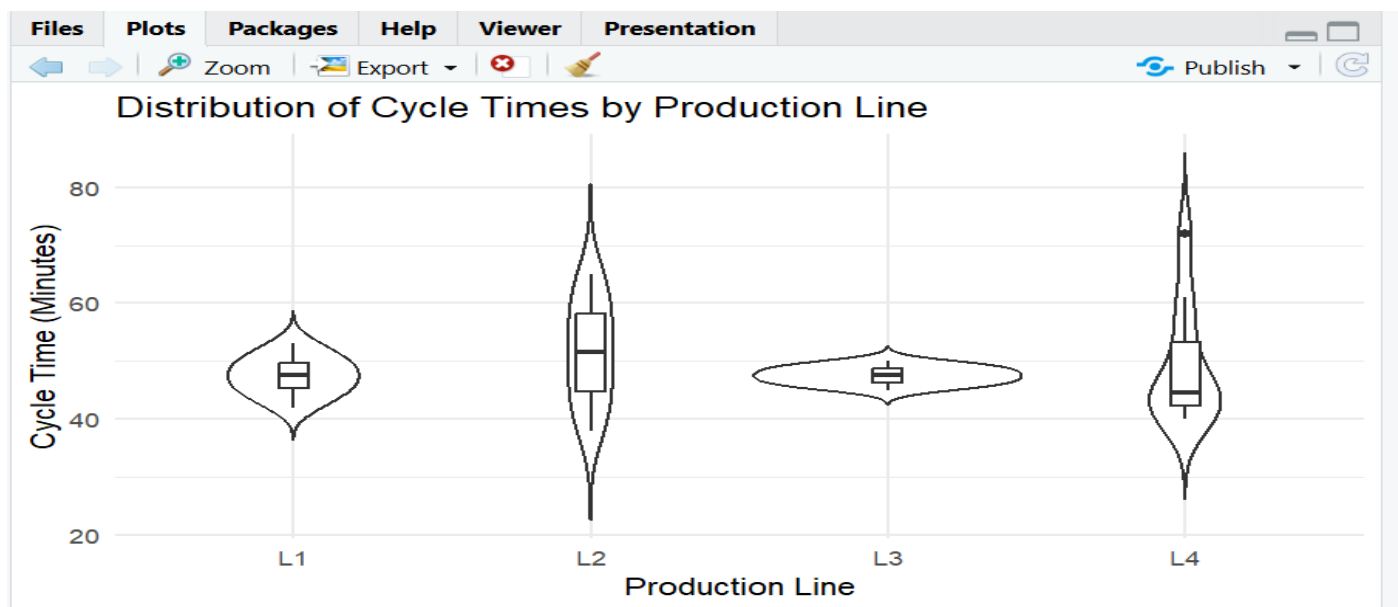
```
)
```



```

ggplot(df2, aes(x = Line, y = Cycle_Time)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Distribution of Cycle Times by Production Line",
    x = "Production Line",
    y = "Cycle Time (Minutes)"
  ) +
  theme_minimal()

```



Greatest Variability

L4 has the greatest variability because its values range from **40 to 72 minutes**, including a long upper spread.

Most Concentrated Distribution

L3 has the most concentrated distribution.

Most L3 values lie between **45 and 50 minutes**, showing low variability.

Why Means Alone Are Insufficient

Two production lines can have similar means but very different:

- Spread
- Variability
- Skewness
- Outliers
- Distribution shape

Therefore, comparing only average cycle time can hide important production consistency problems.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.

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2. Explain which chart better supports comparison of individual proportions and why.

3. Explain why separate pie charts may be less effective for comparing departments.

4. Identify one design choice that could make small proportional differences appear larger.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

```
department <- c("CSE", "ECE", "IT")
```

```
online <- c(52, 45, 30)
```

```
blended <- c(28, 35, 40)
```

```
face_to_face <- c(20, 20, 30)
```

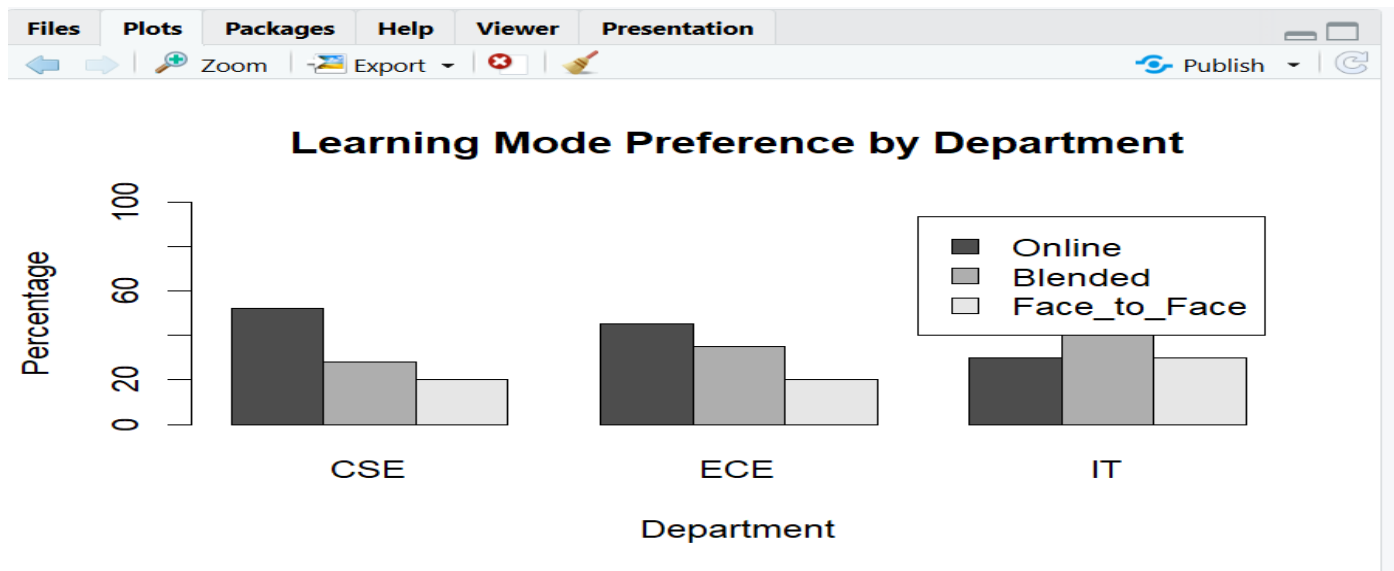
```

learning_data <- rbind(
  Online = online,
  Blended = blended,
  Face_to_Face = face_to_face
)

colnames(learning_data) <- department

barplot(
  learning_data,
  beside = TRUE,
  ylim = c(0, 100),
  main = "Learning Mode Preference by Department",
  xlab = "Department",
  ylab = "Percentage",
  legend.text = TRUE)

```



```

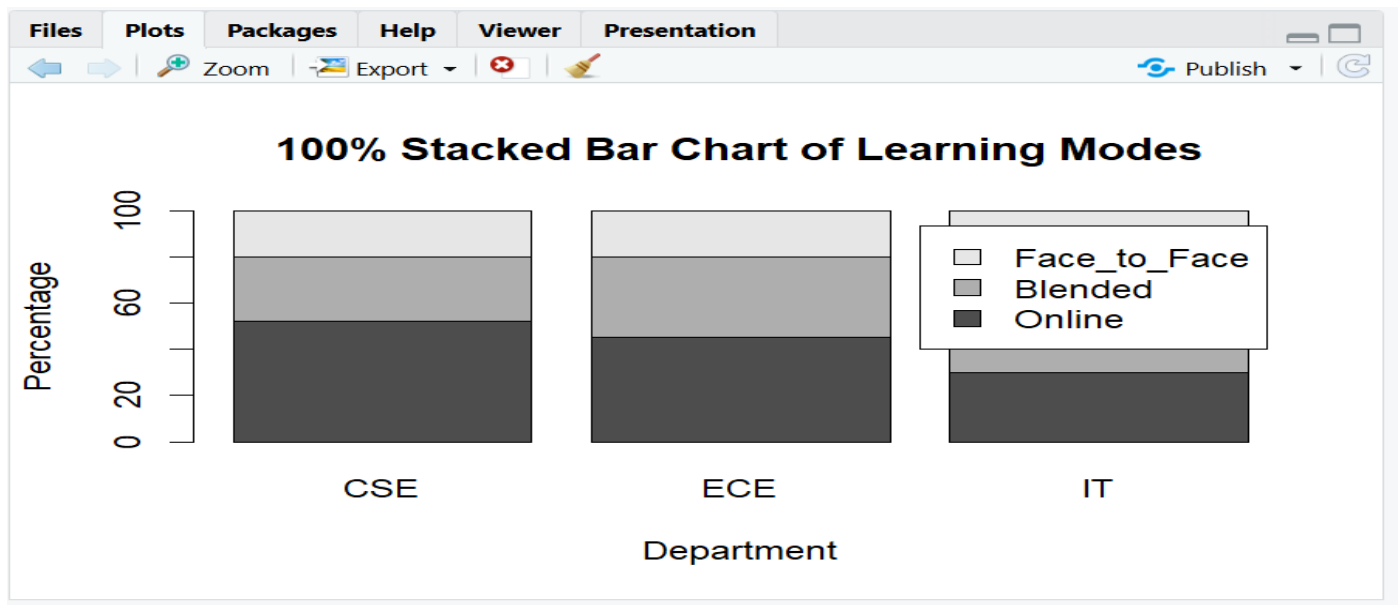
barplot(
  learning_data,

```

```

beside = FALSE,
ylim = c(0, 100),
main = "100% Stacked Bar Chart of Learning Modes",
xlab = "Department",
ylab = "Percentage",
legend.text = TRUE)

```



The **side-by-side bar chart** is better for comparing individual proportions.

For example:

- Online: CSE = 52%, ECE = 45%, IT = 30%
- Blended: CSE = 28%, ECE = 35%, IT = 40%

All bars begin from the same baseline, making differences easier to compare.

The **100% stacked bar chart** is better for showing the overall composition of each department.

Why Pie Charts Are Less Effective

Separate pie charts make comparison difficult because people are less accurate at comparing:

- Angles
- Areas

- Different pie segments

Bar charts provide a common baseline and allow more accurate comparisons.

Misleading Design Choice

A misleading design would be using a **truncated Y-axis**.

For example, starting the Y-axis at 25% instead of 0% could make small differences appear much larger.

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:

1. Construct a treemap showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.
3. Determine the largest category and calculate its percentage of total revenue.
4. Compare treemap and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

Key competency: Treemaps, sunburst charts, nested proportions, area encoding

```
# Install and load packages
```

```
if (!requireNamespace("treemap", quietly=TRUE))
```

```
  install.packages("treemap")
```

```
if (!requireNamespace("plotly", quietly=TRUE))
```

```
  install.packages("plotly")
```

```
library(treemap)
```

```
library(plotly)
```

```
df <- data.frame(
```

```
  Category = c(rep("Electronics",3), rep("Home",3), rep("Fashion",3)),
```

```
  Subcategory = c("Mobiles","Laptops","Accessories",
```

```
                  "Furniture","Appliances","Decor",
```

```
                  "Men","Women","Kids"),
```

```
  Revenue = c(24,18,8,14,10,6,7,8,5)
```

```
)
```

```

print(df)

treemap(df, index=c("Category", "Subcategory"),

        vSize="Revenue",

        title="Revenue Treemap")

tot <- aggregate(Revenue ~ Category, df, sum)

print(tot)

(

data.frame(id=tot$Category, label=tot$Category,

           parent="", value=tot$Revenue),

data.frame(id=paste(df$Category,df$Subcategory,sep="_"),

           label=df$Subcategory, parent=df$Category,

           value=df$Revenue)

)

plot_ly(nodes, ids=~id, labels=~label, parents=~parent,

        values=~value, type="sunburst", branchvalues="total") %>%

layout(title="Revenue Sunburst")

```

```
largest <- tot[which.max(tot$Revenue),]
```

```
total <- sum(df$Revenue)
```

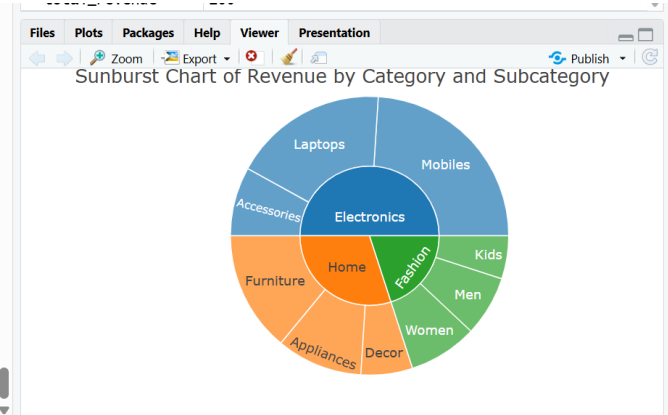
```
cat("\nLargest Category:", largest$Category,
```

```
"\nRevenue:", largest$Revenue, "lakh",
```

```
"\nTotal Revenue:", total, "lakh",
```

```
"\nPercentage:", round(largest$Revenue/total*100,2), "%\n")
```

```
Largest Category: Electronics
>
> cat(
+   "Category Revenue:",
+   largest_category$Revenue,
+   "lakh\n"
+ )
Category Revenue: 50 lakh
>
> cat(
+   "Total Revenue:",
+   total_revenue,
+   "lakh\n"
+ )
Total Revenue: 100 lakh
>
> cat(
+   "Percentage of Total Revenue:",
+   round(percentage, 2),
+   "%\n"
+ )
Percentage of Total Revenue: 50 %
>
```



```
library(treemap)
```

```
df4 <- data.frame(
```

```
Category = c(
```

```
"Electronics",
```

```
"Electronics",
```

```
"Electronics",
```

"Home",

"Home",

"Home",

"Fashion",

"Fashion",

"Fashion"

),

Subcategory = c(

"Mobiles",

"Laptops",

"Accessories",

"Furniture",

"Appliances",

"Decor",

"Men",

"Women",

"Kids"

),

Revenue = c(

24,

18,

8,

14,

10,

6,

7,

8,

5

)

)

print(df4)

treemap(

df4,

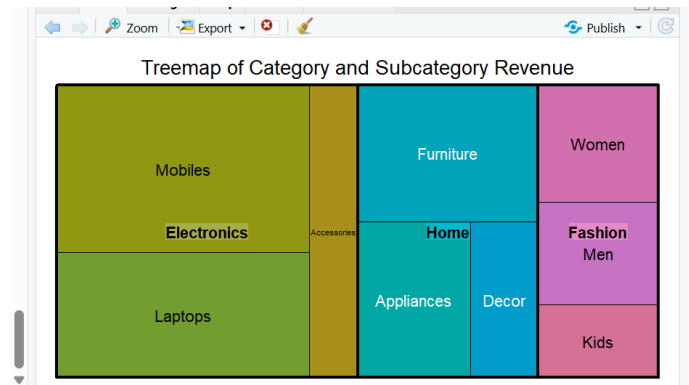
index = c("Category", "Subcategory"),

vSize = "Revenue",

title = "Treemap of Category and Subcategory Revenue"

)

```
> # Display dataset
> print(df4)
  Category Subcategory Revenue
1 Electronics  Mobiles    24
2 Electronics  Laptops    18
3 Electronics Accessories    8
4     Home    Furniture    14
5     Home Appliances    10
6     Home     Decor     6
7 Fashion     Men       7
8 Fashion    Women     8
9 Fashion     Kids     5
>
> # Create Treemap
> treemap(
+   df4,
+   index = c("Category", "Subcategory"),
+   vSize = "Revenue",
+   title = "Treemap of Category and Subcategory Revenue"
+ )
> |
```



Largest Category

Electronics

$24+18+8=50$ lakh

Total revenue:

120 lakh

Percentage:

$$12050 \times 100 = 41.67\%$$

Answer

Electronics is the largest category, contributing 41.67% of the total annual revenue.

Treemap vs Sunburst

Treemap	Sunburst
Uses rectangular areas	Uses circular segments
Easier to compare sizes	Shows hierarchy clearly
Good for identifying largest categories	Good for understanding parent-child relationships
Space-efficient	Can become difficult with many small segments

When Area Encoding Can Mislead

Area encoding can mislead when values are very similar because humans find it difficult to accurately compare areas.

Very small subcategories may also become difficult to see or compare.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:

1. Construct a Sankey diagram showing Origin → Mode → Destination.
2. Construct a parallel-sets representation of the same categorical flows.

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3. Identify the three largest flows and calculate their percentages of all commuters.
4. Compare Sankey and parallel sets for flow magnitude and category relationships.
5. Explain one way poor link widths or category ordering could mislead the viewer.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

```
if (!requireNamespace("plotly", quietly=TRUE))
```

```
  install.packages("plotly")
```

```
library(plotly)
```

```

df <- data.frame(

  Origin=c(rep("Residential",4),rep("Suburban",4),rep("Rural",4)),

  Mode=c("Bus","Bus","Train","Train",

        "Bus","Bus","Train","Train",

        "Bus","Bus","Train","Train"),

  Destination=rep(c("Central","Industrial"),6),

  Commuters=c(35,15,25,10,20,10,30,15,12,8,10,10))

print(df)

om <- aggregate(Commuters~Origin+Mode,df,sum)

md <- aggregate(Commuters~Mode+Destination,df,sum)

nodes <- data.frame(name=c(

  "Residential","Suburban","Rural",

  "Bus","Train","Central","Industrial"

))

links <- rbind(

  data.frame(

```

```

source=match(om$Origin,nodes$name)-1,

target=match(om$Mode,nodes$name)-1,

value=om$Commuters

),

data.frame(

source=match(md$Mode,nodes$name)-1,

target=match(md$Destination,nodes$name)-1,

value=md$Commuters))

plot_ly(

type="sankey",

node=list(label=nodes$name,pad=15,thickness=20),

link=list(source=links$source,

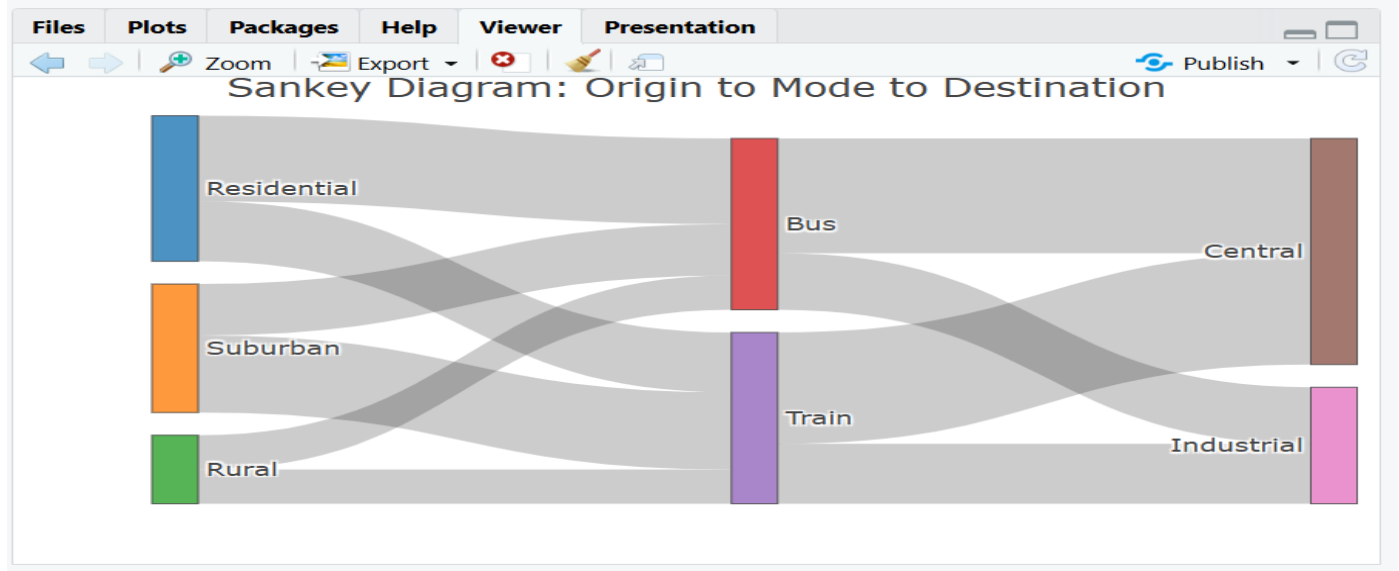
target=links$target,

value=links$value)

) %>%

layout(title="Origin to Mode to Destination")

```



```
ggplot(
  df5,
  aes(
    axis1 = Origin,
    axis2 = Mode,
    axis3 = Destination,
    y = Commuters
  )
) +
```

```
geom_alluvium(  
  
  aes(fill = Mode),  
  
  width = 0.2  
  
) +  
  
geom_stratum(  
  
  width = 0.2  
  
) +  
  
geom_text(  
  
  stat = "stratum",  
  
  aes(label = after_stat(stratum))  
  
) +  
  
scale_x_discrete(  
  
  limits = c(  
  
    "Origin",  
  
    "Mode",  
  
    "Destination"
```

```

),

expand = c(0.1, 0.1)

) +

labs(

title = "Parallel Sets: Origin to Mode to Destination",

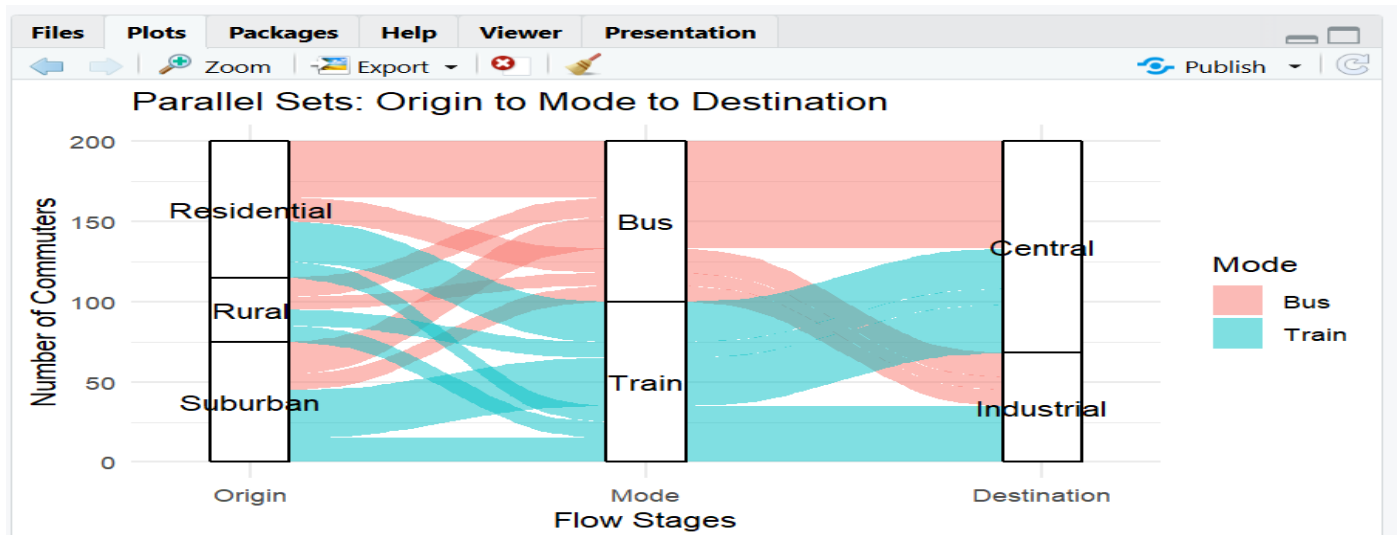
x = "Flow Stages",

y = "Number of Commuters"

) +

theme_minimal()

```



```

total_commuters <- sum(

df5$Commuters

```

```
)

cat(

  "Total Commuters:",

  total_commuters,

  "\n\n"

)

sorted_flows <- df5[

  order(

    -df5$Commuters

  ),

]

top_three <- head(

  sorted_flows,

  3

)

top_three$Percentage <- (
```

```

top_three$Commuters /

total_commuters

) * 100

top_three$Percentage <- round(

top_three$Percentage,

2

)

print(top_three)

>
> # Round percentages
>
> top_three$Percentage <- round(
+   top_three$Percentage,
+   2
+ )
>
>
> # Display result
>
> print(top_three)
      Origin Mode Destination Commuters Percentage
1 Residential Bus      Central      35        17.5
7 Suburban Train      Central      30        15.0
3 Residential Train      Central      25        12.5
>

```

Three Largest Flows

The total number of commuters from the given table is:

$$35+15+25+10+20+10+30+15+12+8+10+10=200$$

1. Residential → Bus → Central

$$20035 \times 100 = 17.5\%$$

2. Suburban → Train → Central

$$20030 \times 100 = 15\%$$

3. Residential → Train → Central

$$20025 \times 100 = 12.5\%$$

Flow	Commuters Percentage	
Residential → Bus → Central	35	17.5%
Suburban → Train → Central	30	15%
Residential → Train → Central	25	12.5%

Sankey vs Parallel Sets

Sankey Diagram

Best for:

- Showing flow magnitude
- Identifying dominant routes

- Finding bottlenecks
- Understanding movement between stages

The width of each connection represents the number of commuters.

Parallel Sets

Best for:

- Comparing relationships among categorical variables
- Observing combinations across multiple categories
- Understanding how categories connect across dimensions

Misleading Design

Poor category ordering can cause links to cross excessively, making the flows difficult to follow.

Incorrect or visually inconsistent link widths could also make a small flow appear more important than a larger flow.

Suggested Evaluation Rubric – Experiment

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.